

Ocean Wave Loads

Potential Flow Theory · Morison Equation · Diffraction Method

Part 1 — Introduction

Why Do Ocean Waves Create Forces?

When a wave rolls past a structure — a wind turbine foundation, a jacket platform— It is accelerating, swirling, reflecting off the surface, and creating pressure fields that change every fraction of a second.

This document will help in understanding and quantifying those forces.

The central question of wave load engineering: given a sea state described by waves of known height, period, and direction — what force does the ocean exert on our structure, and how does that structure respond?

What Makes This Problem Hard?

At first glance, it sounds like a straightforward fluid mechanics problem. In practice, three things complicate it:

- The water surface itself is moving — the boundary condition changes every instant.
- Waves interact with the structure — they reflect, diffract around it, and the structure moves in response to them (for floating structures).
- Real waves are random — not a single clean sinusoid, but a superposition of hundreds of frequency components arriving from different directions.

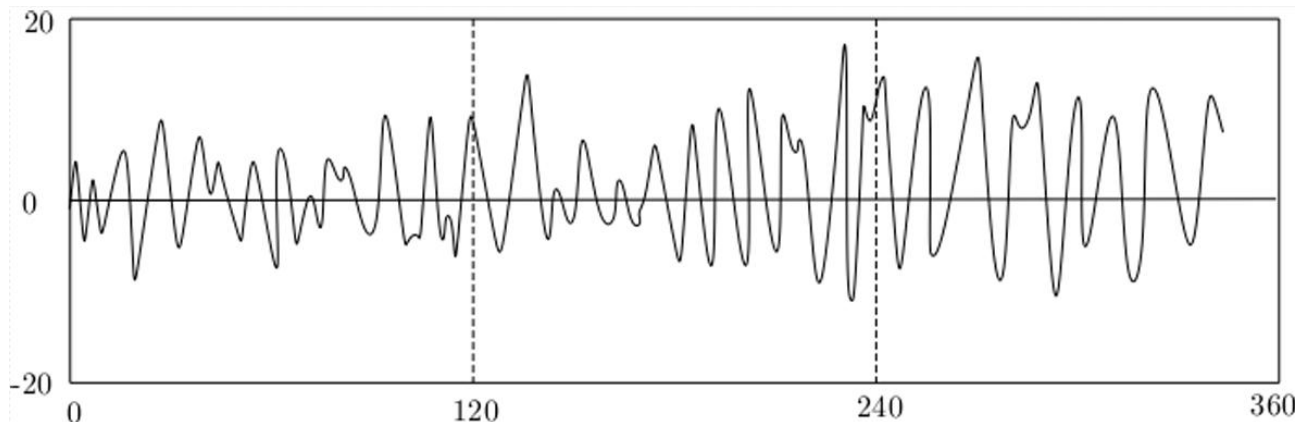


Figure 1: Random wave representing actual sea elevation

Part 2 — Potential Flow Theory

The Simplified Assumption

Real seawater is a viscous, turbulent fluid. Modelling it in its full complexity would require solving the Navier-Stokes equations — a set of nonlinear partial differential equations that have no general closed-form solution and require heavy computing even for simple geometries.

Potential flow theory makes a simplifying assumption: the water is treated as an ideal fluid — one that is inviscid (no viscosity), incompressible (constant density), and irrotational (no spinning or vorticity anywhere in the bulk of the flow). For the vast majority of wave-structure interaction problems — where the flow is not dominated by boundary layer separation — it gives quite accurate results.

The Velocity Potential ϕ

The potential flow is based on a scalar function called the velocity potential, written ϕ (phi). It is defined such that the velocity of water at any point in space and time can be obtained simply by taking its gradient — its rate of change in each direction:

$$u = \partial\phi/\partial x, \quad v = \partial\phi/\partial y, \quad w = \partial\phi/\partial z$$

where u, v, w are the water velocity components in the x, y, z directions

So instead of tracking three separate velocity components everywhere, we only need to find one scalar field ϕ . And because the fluid is incompressible, ϕ must satisfy a — Laplace's equation:

$$\nabla^2 \phi = \partial^2 \phi / \partial x^2 + \partial^2 \phi / \partial y^2 + \partial^2 \phi / \partial z^2 = 0$$

Laplace's equation is one of the most well-studied equations in all of mathematics. We know how to solve it analytically for many geometries, and numerically for almost any geometry.

Linear Wave Theory — Airy Waves

The simplest wave model in offshore engineering is linear wave theory, also known as Airy wave theory (after George Biddell Airy, 1845). It assumes that wave heights are small compared to wavelength — the so-called small amplitude assumption — which linearises the boundary conditions.

Under these assumptions, for a regular sinusoidal wave with height H , period T , and water depth d , the velocity potential has an exact analytical solution:

$$\phi(x, z, t) = (gH / 2\omega) \cdot (\cosh[k(z+d)] / \cosh[kd]) \cdot \sin(kx - \omega t)$$

$$g = 9.81 \text{ m/s}^2 \text{ (gravity)} \quad | \quad \omega = 2\pi/T \text{ (angular frequency)} \quad | \quad k = 2\pi/L \text{ (wave number)} \quad | \quad L = \text{wavelength}$$

From this single expression, everything about the wave field can be derived — water particle velocities, accelerations, pressure at any depth, and ultimately the forces on any structure placed in this wave field.

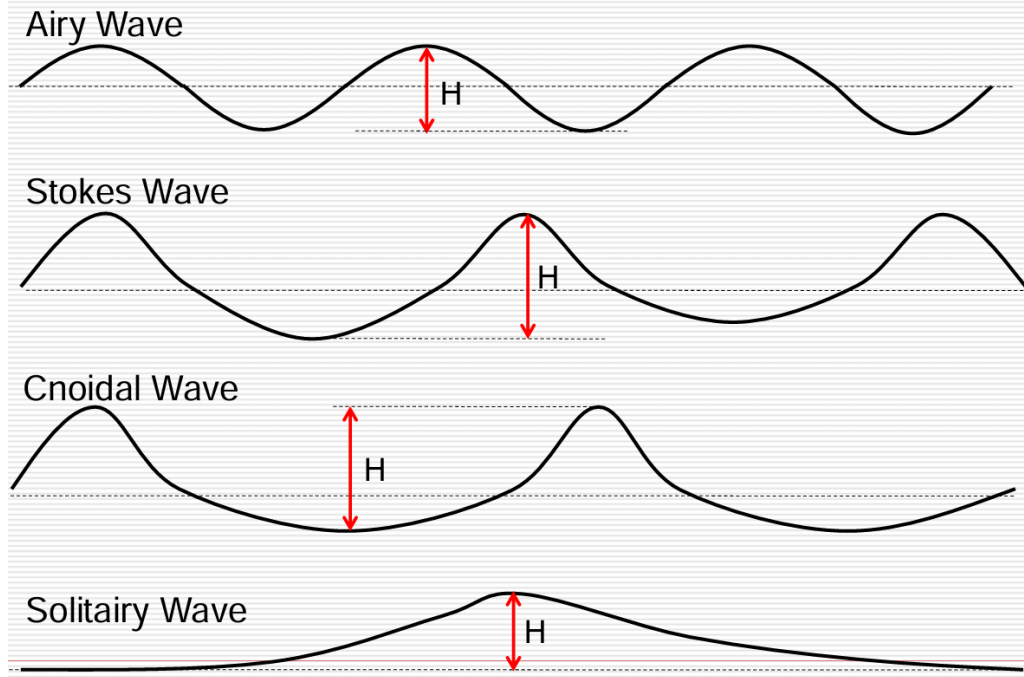


Figure 2: Different Wave forms

Key result from Airy wave theory: the dispersion relation

$$\omega^2 = g \cdot k \cdot \tanh(kd)$$

This links wave frequency to wave number (and hence wavelength). For our example: $\omega = 2\pi/14 = 0.449$ rad/s, giving $L \approx 185$ m

Pressure Field from the Wave

Once we have ϕ , the dynamic pressure anywhere in the water column follows directly from the linearised Bernoulli equation:

$$p(x, z, t) = -\rho \cdot \partial\phi/\partial t = \rho g(H/2) \cdot (\cosh[k(z+d)] / \cosh[kd]) \cdot \cos(kx - \omega t)$$

Notice the $\cosh(k(z+d))/\cosh(kd)$ factor. This tells us how pressure decays with depth. At the surface ($z=0$) it equals 1.0 — full wave pressure. At the seabed ($z=-d$) it equals $1/\cosh(kd)$ — reduced but not zero. For our example with $L=185$ m and $d=30$ m, the seabed sees about 56% of the surface wave pressure. This is why we cannot ignore seabed effects in shallow and intermediate water. However in deep water impact becomes negligible after half of the water depth.

Water Particle Kinematics

The horizontal and vertical velocities of individual water particles are obtained by differentiating ϕ :

$$u(x, z, t) = \partial\phi/\partial x = (\pi H/T) \cdot (\cosh[k(z+d)] / \sinh[kd]) \cdot \cos(kx - \omega t)$$

$$w(x, z, t) = \partial\phi/\partial z = (\pi H/T) \cdot (\sinh[k(z+d)] / \sinh[kd]) \cdot \sin(kx - \omega t)$$

And the horizontal acceleration — which will be crucial for force calculation — is:

$$\partial u/\partial t = (2\pi^2 H/T^2) \cdot (\cosh[k(z+d)] / \sinh[kd]) \cdot \sin(kx - \omega t)$$

Physical intuition: water particles trace circular (deep water) or elliptical (shallow water) paths as a wave passes. It is this accelerating, decelerating motion of the water mass that creates forces on a structure — not just the velocity itself.

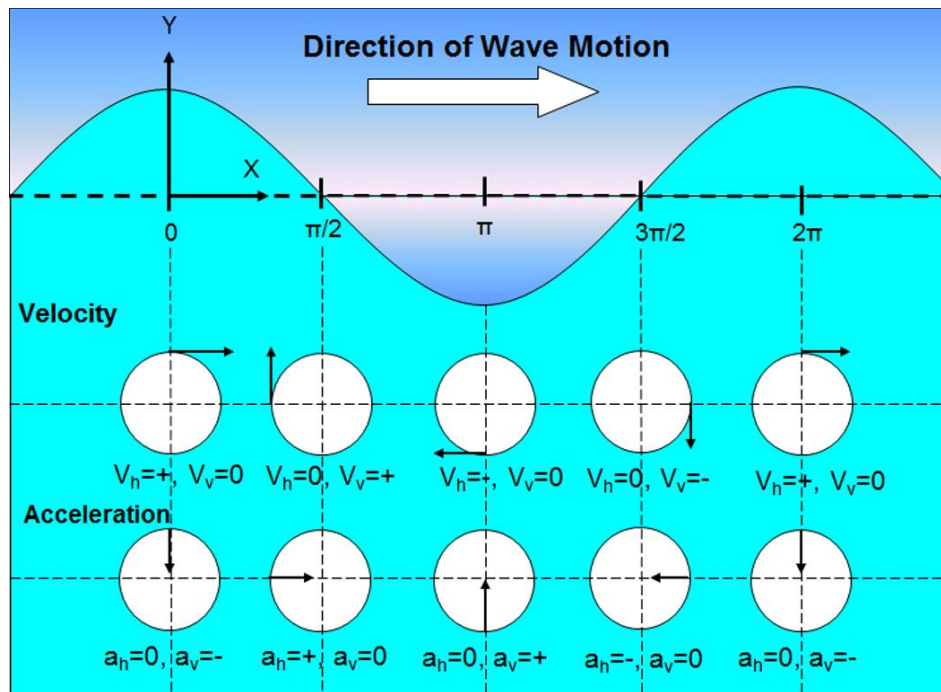


Figure 3: water particle kinematics

Part 3 — Wave and current loads on slender structures

Potential flow theory gives us the pressure field of the undisturbed wave. But when a solid structure is present, two additional physical effects come into play that potential flow alone — in its simplest form — does not capture cleanly for small structures:

- Drag — the water flowing past the structure creates viscous friction and flow separation, pulling on it in the direction of flow.
- Inertia — the accelerating water mass pushes on the structure even if the flow velocity were zero at that instant.

In 1950, J. R. Morison and colleagues proposed a semi-empirical formula that handles both effects for slender cylindrical members. It remains one of the most widely used equations in offshore engineering for structures such as monopile or jackets.

The Morison Equation

Wave and current loading can be calculated by Morison equation as described below:

$$F_T = \left(\frac{1}{2}\right) C_D \rho_w D V |V| + \left(\frac{\pi D^2}{4}\right) C_M \rho_w a$$

Where:

- F_T = total force
- ρ_w = density of water
- C_D = drag coefficient
- C_M = inertia coefficient
- D = diameter of the member including marine growth
- V = water particle velocity
- a = water particle acceleration

The first term in the equation is the drag component (F_D) and the second term is the inertia component (F_I). This can be expressed as:

$$F_T = F_D + F_I$$

Term	Formula	Coefficient	Physical meaning
Inertia term	$\rho \cdot C_m \cdot (\pi D^2/4) \cdot (\partial u/\partial t)$	$C_m \approx 2.0$ for circular cylinder	Force from the accelerating water mass. Proportional to acceleration, not velocity. Dominant for large structures.
Drag term	$(1/2) \cdot \rho \cdot C_d \cdot D \cdot u \cdot u $	$C_d \approx 0.7-1.2$ for circular cylinder	Force from viscous drag and flow separation. Proportional to velocity squared. Dominant for slender structures.

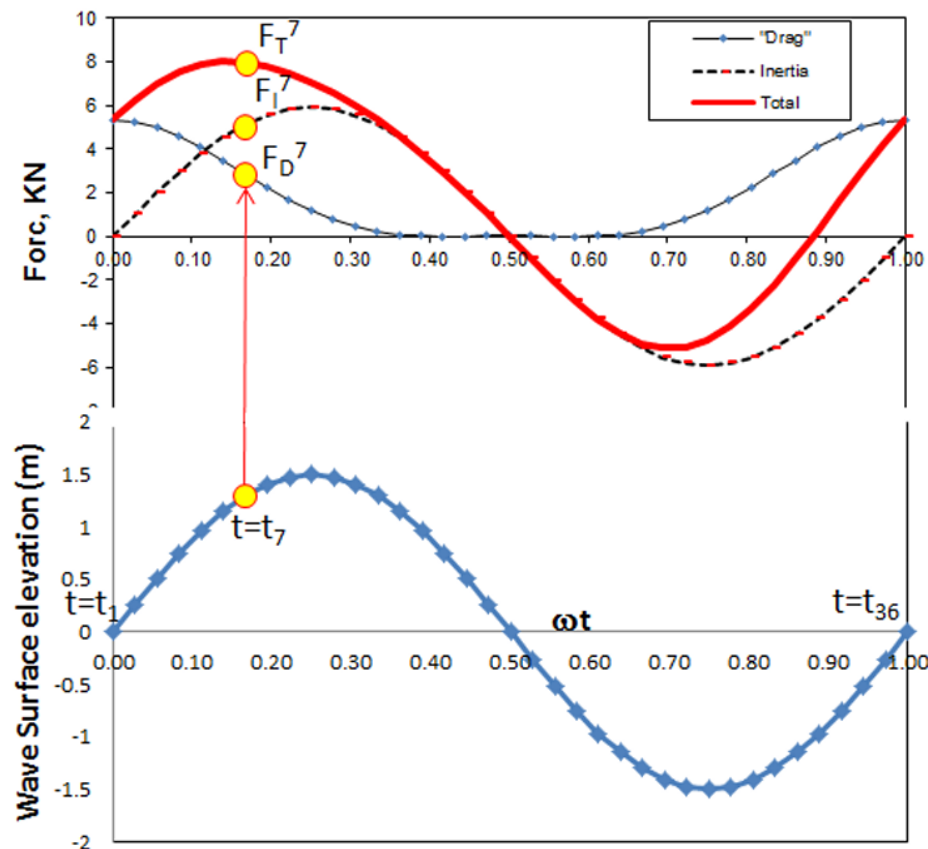


Figure 4: Total force as function of surface elevation

Figure 4 illustrates the time-dependent nature of wave loading on a marine structure over one complete wave cycle. The lower plot shows the variation of wave surface elevation with time, while the upper plot presents the corresponding drag force, inertia force, and total hydrodynamic force calculated using the Morison equation.

It can be observed that the hydrodynamic loading continuously changes with wave phase. During the wave crest region, the water particle motion acts in one direction, producing positive force values, whereas during the wave trough region the particle motion reverses, resulting in forces acting in the opposite direction. The total force at any instant is therefore dependent on both the instantaneous water particle velocity and acceleration.

When Can You Use Morison?

The Morison equation was derived for structures that are small relative to the wavelength. The rule of thumb is the Keulegan-Carpenter number KC and the diffraction parameter:

Diffraction parameter: $D/L < 0.2 \rightarrow$ Morison equation is valid

Keulegan-Carpenter number: $KC = \pi \cdot H / D$ (higher $KC \rightarrow$ drag dominates)

For our North Sea example: $D = 6$ m, $L = 185$ m, so $D/L = 6/185 = 0.032$. This is well below 0.2, so the Morison equation is applicable. However, as monopiles get larger ($D = 8\text{--}10$ m is now common), D/L approaches 0.1 and diffraction effects become significant — which leads us to the next method.

Integrating the Morison Equation — Total Force

To get the total horizontal force on the monopile, we integrate the Morison force per unit length from the seabed to the water surface. At the instant of maximum inertia force (when the wave crest is about a quarter wavelength away from the pile), the inertia term dominates:

$$F_{\text{inertia}}(t) = \int_{-d}^0 \rho \cdot C_m \cdot (\pi D^2 / 4) \cdot (\partial u / \partial t) \, dz$$

For a linear Airy wave, this integral has a closed-form solution. The maximum inertia force on a vertical pile is:

$$F_{\text{max}} = \rho \cdot C_m \cdot (\pi D^2 / 4) \cdot (2\pi^2 H / T^2) \cdot (\tanh(kd) / k)$$

Figure 5 illustrates the application of Morison wave loading on a vertical cylindrical member subjected to wave action. The incident wave profile changes with time, as shown by the wave elevations at $(t = t_1)$ and $(t = t_2)$, demonstrating the time-dependent nature of the hydrodynamic loading. As the wave crest and trough propagate, the direction and magnitude of water particle kinematics vary continuously, causing the hydrodynamic forces on the cylinder to also change with time.

The cylinder is discretized into multiple segments along the water depth, and the local hydrodynamic force acting on each segment $((F_{\{n-1\}}), (F_n), (F_{\{n+1\}}))$ is calculated based on the local water particle velocity and acceleration. The total horizontal force $((F_T))$ on the structure is then obtained by integrating the distributed load over the submerged length from the free surface to the seabed depth $((0)$ to $(-h))$.

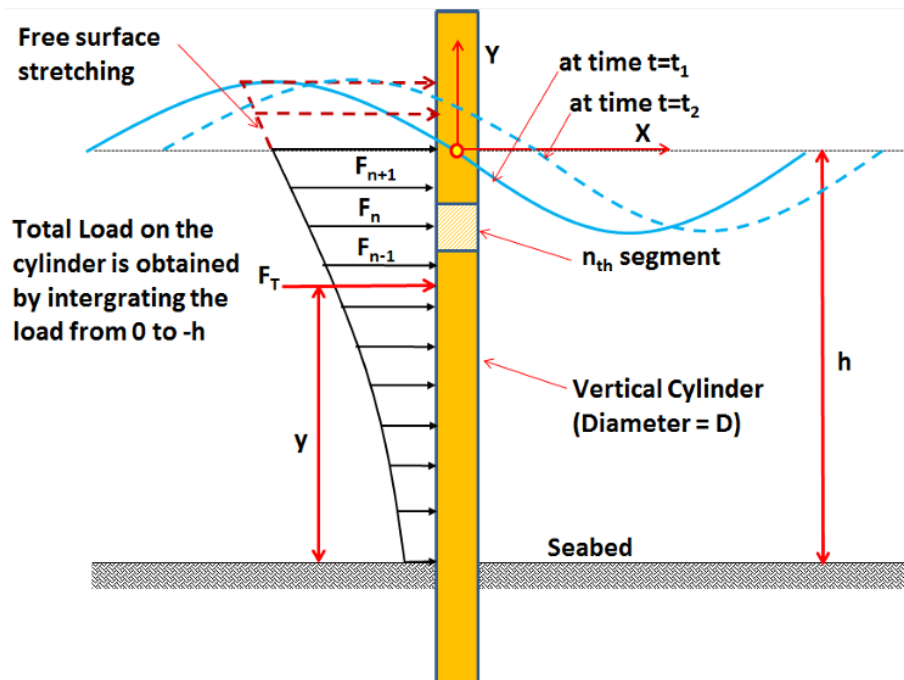
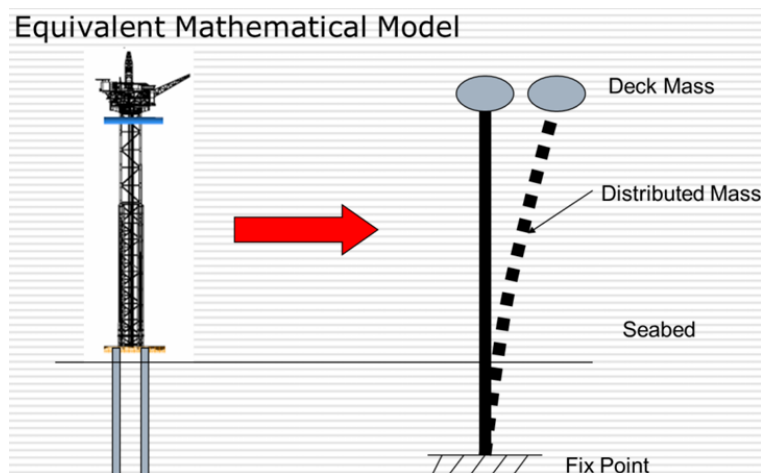


Figure 5: Time-dependent Morison wave loading on a vertical cylinder

Worked out Example – Monopile

GitHub Link : [Wave Load calculation](#)

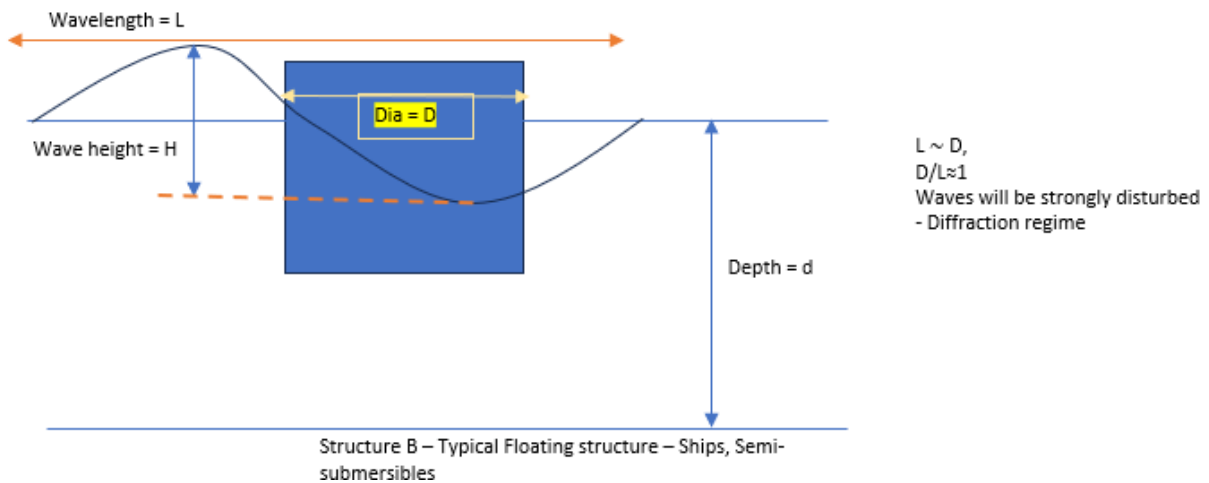
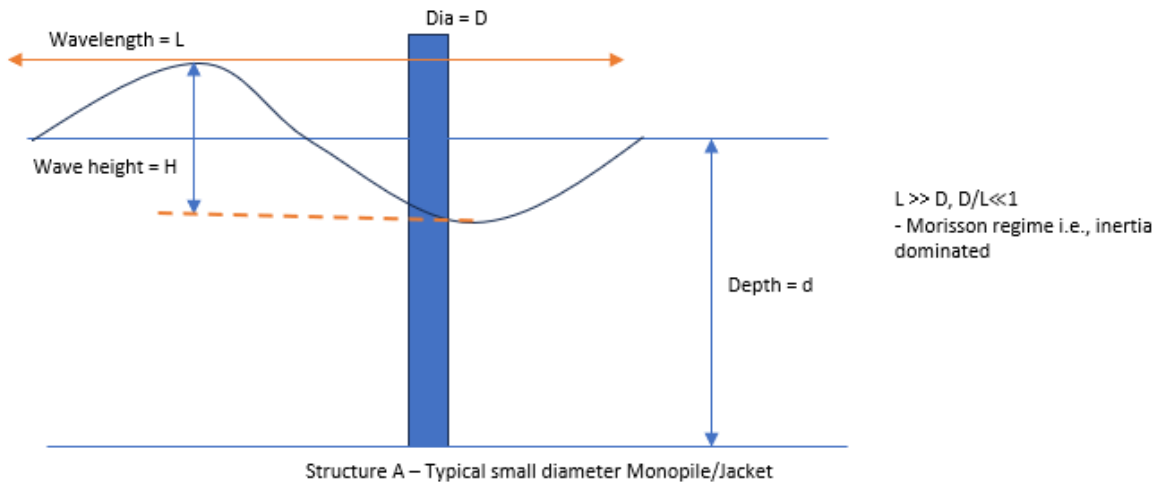
Here, in above example it is assumed that pile is rigid (no deflection). When response is assumed then system can be solved using Spring Mass damper system as shown below



Part 4 — Diffraction Theory

When the Structure Is Too Large to Ignore

The Morison equation treats the structure as if it were not there — it uses the undisturbed wave kinematics and then applies empirical coefficients to account for the structure's effect. This works well when the structure is small compared to the wavelength. But what happens when it is not?



Imagine holding your hand in a garden hose stream versus placing a football in its path. With your hand (small), you feel a force roughly proportional to the stream's velocity. But the football (large) fundamentally redirects the entire stream — the flow pattern around it is nothing like the undisturbed stream. The same thing happens with large offshore structures and ocean waves.

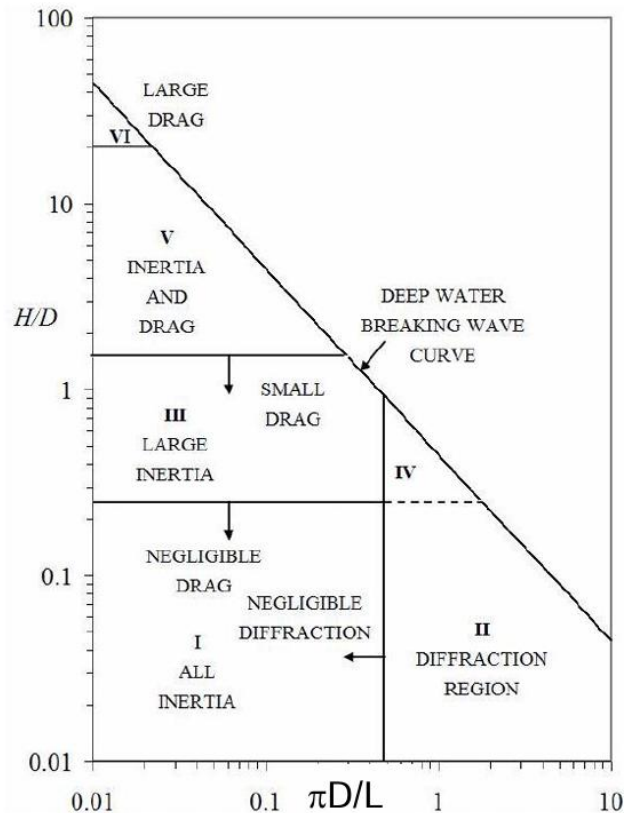


Figure 6: Force Regime

When $D/L > 0.2$, the structure scatters (diffracts) the incident waves. The scattered waves interact with the incident waves, creating a complex pressure field that the simple Morison equation cannot capture. Diffraction analysis is required.

Setting Up the Diffraction Problem

In diffraction theory, the total velocity potential is split into three components:

$$\varphi_{\text{total}} = \varphi_{\text{incident}} + \varphi_{\text{scattered}} + \varphi_{\text{radiation}}$$

$\varphi_{\text{incident}}$ = the undisturbed incoming wave

$\varphi_{\text{scattered}}$ = waves reflected/diffracted by the fixed structure

$\varphi_{\text{radiation}}$ = waves generated by the structure's own motion (for floating bodies)

For a fixed structure like our monopile, the radiation term is zero. We only need the incident and scattered potentials. The scattered potential must satisfy:

- Laplace's equation throughout the fluid domain (same as before)
- The free surface boundary condition — pressure equals zero at the wave surface
- The seabed boundary condition — no flow through the bottom
- The body boundary condition — no flow through the structure's surface

- The radiation condition — scattered waves propagate outward and die away at infinity

This is a well-posed mathematical problem. For simple geometries like a vertical circular cylinder, it has an exact analytical solution. For complex geometries (a full jacket platform with dozens of members), it is solved numerically using Finite element methods or finite volume methods.

The Physical Mechanism of Diffraction

Why does diffraction reduce the force? Because as the structure gets larger, it starts to scatter the incident wave. The scattered wave radiates outward, carrying energy away from the structure. The wave on the downstream side of the cylinder is partially sheltered — so the net pressure difference from front to back is less than the undisturbed wave pressure would suggest.

For very large structures ($kR \gg 1$), the cylinder is large enough to reflect most of the incident wave energy back toward the wave source. The force then becomes dominated by the reflected wave pressure rather than the inertia of accelerating water — a completely different physical regime from the Morison case.

Boundary Element Method — Numerical Diffraction

For real offshore structures — semi-submersibles, ship hulls — the geometry is far too complex for an analytical solution. Numerical diffraction analysis uses the Boundary Element Method (BEM), also called the panel method or 3D diffraction analysis.

The concept is elegant: instead of solving Laplace's equation throughout the entire fluid volume (which would be enormous), BEM exploits Green's theorem to convert the volume problem into a surface integral. Only the structure's wetted surface needs to be discretised into panels — typically a few hundred to a few thousand for a full-scale platform.

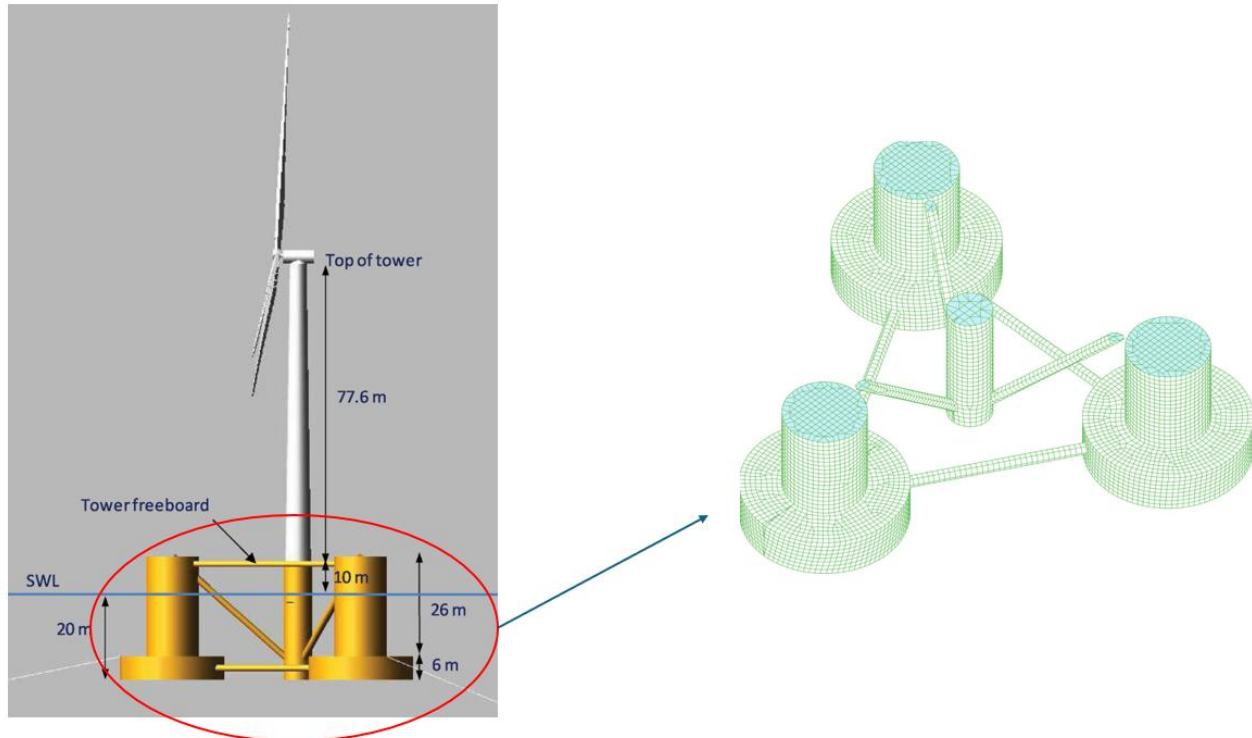


Figure 7: Meshed DeepCWind Semi submersible Platform (Zabala et al. 2026)

1. The wetted surface of the structure is divided into flat quadrilateral or triangular panels.
2. On each panel, the unknown scattered wave potential is assumed constant (constant-strength source distribution).
3. For each panel, the influence of every other panel's source distribution on it is computed — this gives a dense $N \times N$ matrix equation.
4. The matrix is solved to find the source strengths that satisfy the no-flow-through boundary condition on every panel.
5. Once the source strengths are known, the pressure on every panel is computed from Bernoulli's equation.
6. Integrating the pressure over all panels gives the total wave force — including all diffraction and radiation effects.

Part 5 — From Wave Loads to Structural Response

The Monopile Is Not Rigid

Everything so far has assumed the monopile is perfectly rigid — it does not move in response to the wave force. For most static analyses, this is adequate. But for dynamic analyses — fatigue, resonance, operational loading during turbine operation — the flexibility of the structure matters enormously.

A 6 m diameter, 80 mm wall monopile in 30 m water has a natural frequency (its 'resonant frequency', like the pitch of a tuning fork) of approximately 0.25–0.35 Hz. This is:

- Higher than wave frequencies (0.05–0.12 Hz for North Sea storms) — so waves do not directly excite resonance
- Close to 3P frequency from the turbine rotor (typically 0.3–0.5 Hz for modern turbines) — this is a critical design constraint
- Close to 1P frequency from rotor imbalance — another source of structural excitation

The key design constraint for monopile natural frequency is to stay in the 'soft-stiff' window: above 1P (rotor once-per-revolution) and below 3P (blade-passing frequency). Too stiff and the pile is uneconomical; too soft and it resonates with waves. Statistical wave load analysis feeds directly into this dynamic design problem.

The Dynamic Amplification Factor

When a structure is loaded at a frequency close to (but not at) its natural frequency, the response is amplified. This amplification is quantified by the Dynamic Amplification Factor (DAF):

$$DAF = 1 / \sqrt{[1 - (f/f_n)^2]^2 + (2 \cdot \zeta \cdot f/f_n)^2}$$

f = loading frequency | f_n = natural frequency | ζ = damping ratio (typically 0.01–0.05 for steel offshore structures)

For our monopile with f_n ≈ 0.30 Hz and dominant wave frequency f ≈ 0.07 Hz: f/f_n ≈ 0.23, giving DAF ≈ 1.06 — less than 6% amplification. The structure is well clear of resonance for normal waves. But during startup and shutdown, the turbine passes through the 1P and 3P frequencies — and the DAF can be much higher for those transient loads.

Fatigue

A single extreme wave creates a large force but does not by itself break a monopile designed to survive it. What eventually determines the structural lifetime is fatigue — the accumulation of damage from millions of smaller wave cycles over 25 years.

The industry uses SN curves (stress vs number of cycles to failure) in combination with the wave load distribution from statistical analysis. The connection back to our earlier statistical modelling is direct:

7. The wave statistics (spectrum, distribution of H and T) give the distribution of wave loads at each frequency.
8. Each load amplitude creates a stress cycle in the steel. Larger waves create larger stresses.
9. The SN curve tells us how many cycles of each stress amplitude the steel can survive.

10. Miner's rule sums the fractional damage from each cycle: $D = \sum(n/N)$ where n is actual cycles and N is allowable cycles at that stress level.
11. Design life requires $D < 1.0$ with appropriate safety factors (typically $D < 0.33$ for monopiles).

This is why the wave load calculation is not just a one-time static analysis. It feeds into a probabilistic, frequency-domain fatigue calculation that accounts for every sea state the structure will experience over its lifetime — from gentle 0.5 m waves on calm days to 18 m storm waves on winter nights.

Part 6 — Summary and Key Takeaways

Summary

Method	What it does	When to use
Potential Flow Theory	The mathematical framework — treats water as ideal fluid, derives wave kinematics from Laplace's equation and velocity potential ϕ	Foundation for all three methods
Morison Equation	Semi-empirical: adds drag (C_d) and inertia (C_m) forces on slender members using undisturbed wave kinematics	Use when $D/L < 0.2$
Diffraction Analysis	Solves for the scattered wave field created by the structure itself; exact (MacCamy-Fuchs) or numerical (BEM panel method)	Use when $D/L > 0.2$ or for floating bodies

What Engineers Actually Do

In a real offshore wind project, the wave load workflow typically looks like this:

1. Collect metocean data — years of wave measurements from buoys or hindcast models.
2. Characterise sea states statistically — fit a scatter diagram of H_s (significant wave height) vs T_p (peak period).
3. Run frequency-domain fatigue analysis — for each sea state bin, compute the wave load spectrum, apply DAF, integrate damage.
4. Run extreme load analysis — select the 50-year and 100-year design waves, compute peak force, check strength.
5. Run time-domain simulations — for the governing cases, run coupled aero-hydro-elastic models (FAST, OrcaFlex, SESAM) to capture nonlinear effects.
6. Iterate on the structural design — wall thickness, embedment depth, grout connection — until all limit states are satisfied.

Key References & Further Reading

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